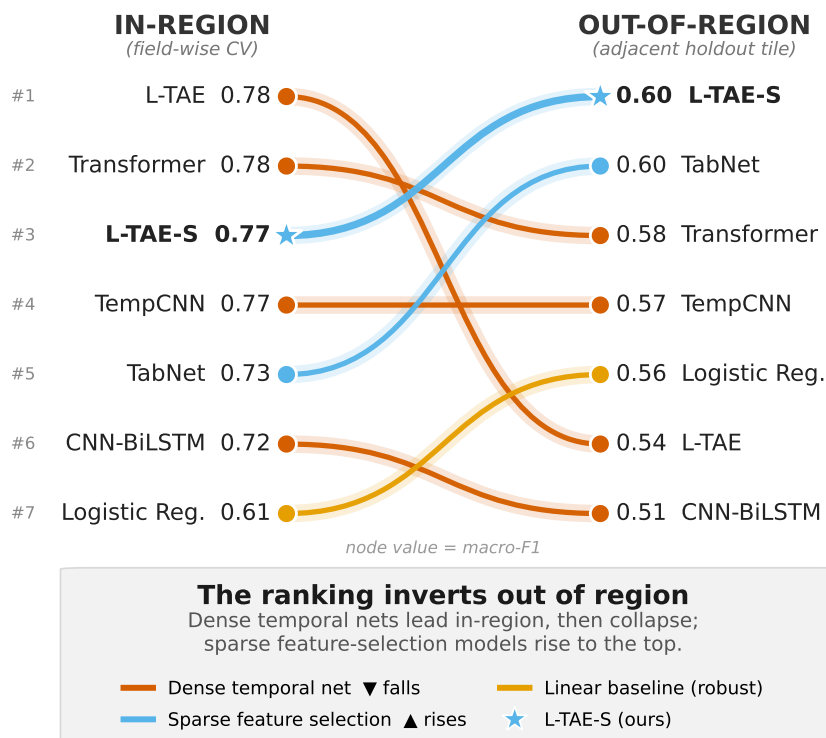


1 Graphical Abstract

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Sparse Feature Selection Outweighs Deep Learning Capacity for Out-of-Region Transfer in Crop Classification

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Abstract

Crop-type maps from satellite image time series are a backbone of agricultural monitoring, yet the machine-learning models that produce them are almost always trained and tested within a single region, leaving open how they behave when deployed to new areas where no labels were collected. Using a South African Sentinel-2 dataset (optical only), we benchmark twenty-three classical machine-learning and deep-learning models under out-of-region transfer, applying them without any target-region labels or adaptation to an adjacent, non-overlapping holdout tile, and identify the model property most associated with success: sparse, axis-aligned feature selection. Models that commit to a few features at a time, gradient-boosted trees, logistic regression, and the attentive TabNet, transfer robustly (macro-F1 0.56–0.60), whereas high-capacity dense temporal networks (CNN-BiLSTM, a Transformer encoder, L-TAE) degrade sharply; in a controlled comparison, adding sparse feature selection to L-TAE improves transfer more than adding attention capacity does. We turn this insight into a method: L-TAE-S, a sparse-attention temporal encoder that ties the best model for out-of-region macro-F1 (0.60) at a fraction of the training cost, with a field-level variant that matches this accuracy while training in seconds per seed. We further show that conventional in-region validation inverts these rankings, the dense networks that lead in-region (macro-F1 0.72–0.78) fall furthest under transfer, which is why models intended for operational crop mapping should be evaluated outside the region they were trained on. Field-level aggregation is a second, substitutable robustness lever, and the most transferable models retain near-full accuracy using only a quarter of the training fields. Together these results give practitioners a concrete design principle for crop classifiers that must be applied to nearby unlabeled farmland.

Keywords: Crop classification, deep learning, feature selection, remote sensing, Sentinel-2, out-of-region transfer, time-series classification

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1. Introduction

Accurate and timely crop classification is foundational to precision agriculture and to the agencies that monitor food security and agricultural land use, enabling yield forecasting, resource optimization, and food-security assessment (Bégué et al., 2020; Ienco et al., 2017). The Sentinel-2 constellation, with its multispectral richness and roughly five-day revisit, has made dense multi-temporal optical imagery widely available. Yet reliable crop classification remains hard throughout Africa and the Global South: spectral similarity between crops, irregular planting and phenological cycles, persistent cloud cover, and rain-fed dryland management all degrade accuracy (Bégué et al., 2020; Mahlayeye et al., 2024). Classical methods such as Random Forest (RF) and XGBoost have proven effective and efficient on engineered features (Mahlayeye et al., 2024; Mann et al., 2026; Mann and Warner, 2017; Mann et al., 2019), but their performance hinges on hand-crafted feature engineering (Maxwell et al., 2018; Belgiu and Drăguț, 2016). Deep learning, recurrent and convolutional temporal models, reduces reliance on hand-crafted predictors and captures phenological structure that summary statistics miss (Zhu et al., 2017; Rußwurm and Körner, 2018; Pelletier et al., 2019), but its generalizability in complex, low-label settings remains uncertain (Pankajakshan et al., 2025).

A second, and for our purposes critical, gap concerns *how generalization is measured*. Much of the crop-classification literature reports accuracy under random or field-wise cross-validation drawn from a single region (Belgiu and Drăguț, 2016; Garnot and Landrieu, 2020; Pelletier et al., 2019; Rußwurm and Körner, 2018; Ienco et al., 2017). Such protocols hold out individual pixels or fields but not *space*: training and test observations share the same atmosphere, soils, management calendars, and acquisition geometry, so the reported numbers reflect in-distribution fit rather than transfer to new terrain. Because operational maps are produced by applying a model to regions where no labels were collected, this distinction matters, yet studies that re-rank models under a spatially disjoint holdout remain rare (Pankajakshan et al., 2025; Jin et al., 2019; Rustowicz et al., 2019). This differs from the bulk of the transfer-learning literature, which adapts a model using target-domain data: labels, unlabeled imagery, or self-supervised pre-training (Ma et al., 2024; Wang et al., 2023; Mai et al., 2025). Our setting is the no-target-access case (domain generalization): models are trained on source tiles and applied to the holdout directly, with no adaptation, so what determines transfer is a property intrinsic to the model rather than a target-specific tuning step.

The Western Cape of South Africa concentrates the operational challenges, smaller fields, rain-fed dryland systems, heterogeneous planting rotations, and persistent winter cloud cover, while offering an unusually rich, high-quality labeled dataset (Radiant Earth Foundation, 2024). This makes it a realistic test bed: rich enough to train and compare modern architectures fairly, yet hard enough to expose the failure modes that matter in practice. We evaluate deep temporal models (CNN-BiLSTM, TabNet, L-TAE, a Transformer encoder, and TempCNN) and deep patch-based 2D/3D CNNs against classical XGBoost,

59 LightGBM, and RF. Critically, we evaluate every model twice: under field-wise
60 cross-validation *within* the training region, and on an adjacent, non-overlapping
61 holdout tile that no training pixel touches. We restrict inputs to multi-temporal
62 Sentinel-2 optical imagery alone, to isolate the spectral-time-series signal and
63 maximize applicability where only optical data are available. All code is public
64 (Mann, 2025a).

65 This study makes five contributions. First, a multi-architecture benchmark
66 (twenty-three classical, deep temporal, and deep patch-based models) showing
67 that in-region versus out-of-region transfer evaluation produce *inverted* rankings
68 on the same data. Second, empirical evidence that models with sparse, axis-
69 aligned feature-selection, the property behind two decades of RF and gradient-
70 boosted-tree dominance, are more stable when moved to a holdout tile, and
71 TabNet inherits it via sparsemax masks. Third, a controlled architectural compar-
72 ison, tested with paired bootstraps, in which adding sparse feature selec-
73 tion improves transfer more than adding attention capacity. We introduce L-
74 TAE-S, a sparse-attention encoder that ties TabNet for the best out-of-region
75 macro-F1, with a field-level variant that matches this accuracy while training
76 in seconds per seed. Fourth, identification of field-level aggregation as a second,
77 substitutable robustness lever that recovers transfer for dense models but is re-
78 dundant for sparse ones. Fifth, a data-efficiency analysis under out-of-region
79 transfer showing the most transferable models are also the most data-efficient,
80 holding near-full holdout accuracy on a quarter of the training fields; as a sup-
81 porting experiment, a controlled comparison of `xr.fresh` time-series features
82 against raw reflectance for XGBoost and TabNet under out-of-region trans-
83 fer. This leads to our central question: across classical machine-learning and
84 deep-learning models, which most reliably classifies crops from multi-temporal
85 Sentinel-2 optical imagery when applied, without target-region labels, to an ad-
86 jacent region outside the training tiles, and what model properties determine
87 whether in-distribution accuracy survives the transfer? Our aim is not to build
88 a single operational model but to test how model families behave under the low-
89 information conditions common across Africa: no target-region labels, a modest
90 single-season training set, and optical imagery only.

91 1.1. Related Work

92 Early crop classification relied on classical learners (RF, SVM, logistic regres-
93 sion, gradient-boosted trees) operating on engineered descriptors (spectral in-
94 dices, temporal summaries) valued for efficiency, interpretability, and robustness
95 on structured data (Maxwell et al., 2018; Belgiu and Drăguț, 2016; Breiman,
96 2001; Chen and Guestrin, 2016; Ke et al., 2017), though their performance
97 depends on feature engineering that can limit transferability across agroeco-
98 logical regions (Wardlow et al., 2007; Mahlayeye et al., 2024). Deep models
99 learn representations directly from reflectance sequences: CNNs capture local
100 spectral-spatial pattern (Kussul et al., 2017), and recurrent and attention ar-
101 chitectures model seasonal phenology (Ienco et al., 2017; Rußwurm and Körner,
102 2018; Garnot and Landrieu, 2020; Rußwurm and Körner, 2020). Other designs
103 combine convolutional feature extraction with temporal modeling (Fan et al.,

2023; Tufail et al., 2025; Bandar and Coşkunçay, 2024), or use deep backbones as feature extractors feeding classical classifiers (Espejo-Garcia et al., 2020; Koklu et al., 2022; Yang et al., 2020). A recurring assumption is that richer learned representations generalize better than compact engineered ones; whether this holds under out-of-region transfer, rather than within-region validation, is less established and is a central question here. Architectures that explicitly target cross-scenario generalization (e.g., Cropformer (Wang et al., 2023)) remain the exception and are rarely benchmarked against classical and deep baselines on a common spatially disjoint holdout. We fill this gap with a systematic comparison under *both* protocols, and additionally evaluate `xr_fresh` as an automated time-series feature-extraction framework in a crop-classification setting (Mann, 2025b).

2. Data

The study uses a labeled field dataset distributed through Radiant Earth’s MLHub (Radiant Earth Foundation, 2024) and used as the basis of the AI4FoodSecurity (South Africa) challenge (AI4EO, 2021), in an agriculturally productive part of the Western Cape (Fig. 1). Two adjacent Sentinel-2 tiles (`34S_19E_258N`, `34S_19E_259N`) form the training region (4,150 labeled fields over 622 km²; mean field 15.0 ha); a third, immediately adjacent tile (`34S_20E_259N`) is reserved as a holdout for out-of-region transfer evaluation (2,417 fields over 282 km²). The tiles do not overlap, so no pixel or field is shared, and transfer must contend with different soils, geometries, and acquisition conditions. Because the holdout lies within the same agroecological zone and growing calendar, this is a relatively *local* shift, and the gaps we report are best read as a conservative lower bound on degradation under broader transfer. Throughout, “in-region” denotes a field-wise train/validation/test split within the training region; “out-of-region” (OOR) or “holdout” denotes predictions on the adjacent, non-overlapping tile, made without any target-region labels or adaptation. We use “out-of-region” rather than the broader “spatial transfer” because the holdout tests a local shift, not transfer across agroecological zones or years.

We classified five crop types: wheat, canola, lucerne/medics (alfalfa), barley, and small-grain grazing. The classes are heavily imbalanced; lucerne/medics is 43% of training fields and 56% of the holdout (Supplementary Material, Fig. S1(a)), which motivates macro-F1 as the primary metric (Sec. 3.3) and class-imbalance handling during training. Spectrally, wheat and barley overlap strongly, the dominant source of persistent confusion (Supplementary Material, Fig. S2(b)). Sentinel-2 Level-2A surface reflectance was acquired for the January–December 2017 growing season, cloud-masked with `s2cloudless` plus a shadow mask, and composited monthly. Six features per month are used: bands B2, B6, B11, B12, plus EVI and Hue, with EVI and Hue computed per scene before compositing, chosen to span the most informative spectral regions while limiting redundancy among correlated bands (Mann et al., 2026; Tufail et al., 2025). June 2017 yielded no usable imagery and May was discarded for cloud, leaving 10 monthly observations, so each pixel is a 60-dimensional

Study area: Western Cape, South Africa

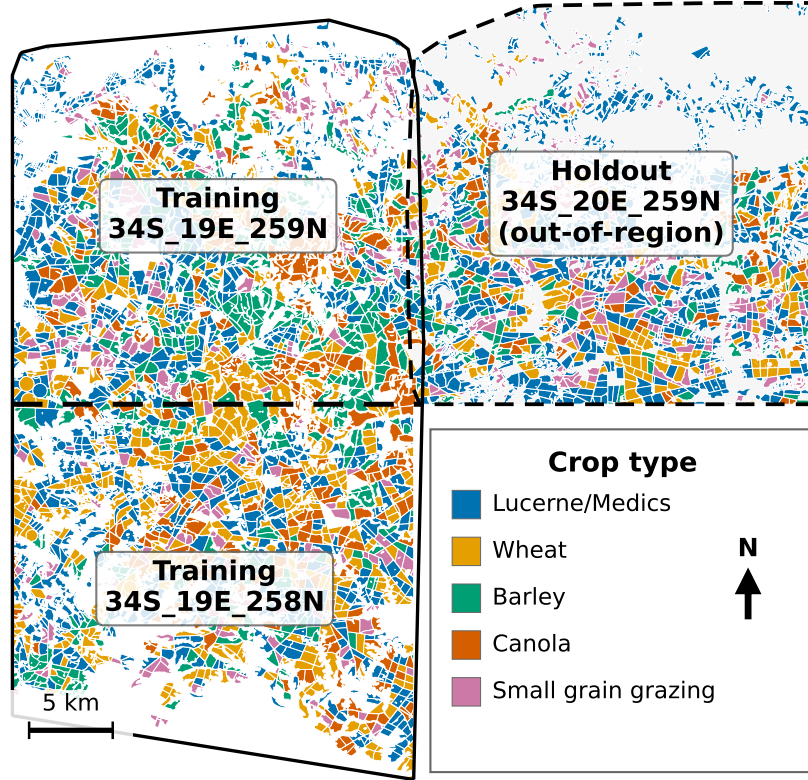


Figure 1: Study area in the Western Cape, South Africa. Two adjacent Sentinel-2 tiles (34S_19E_258N, 34S_19E_259N) form the training region; the immediately adjacent 34S_20E_259N tile (dashed) is held out for out-of-region transfer evaluation. Fields are colored by crop type; the tiles are spatially disjoint, so no pixel or field is shared between training and holdout.

time series (6 features $\times$ 10 months). Field polygons were digitized from sub-meter aerial photography and carry a single seasonal crop label (Radiant Earth Foundation, 2024); all predictions are made at the field level.

3. Methodology

We compare two model families: classical machine learning on engineered `xr_fresh` features (9 models) and deep learning (14), the latter comprising deep temporal models on per-pixel or per-field time series (12) and deep patch-based models on image patches (2). To prevent leakage, all train/validation/test splitting is performed at the field level with a fixed random seed, so no field's pixels

straddle partitions; pixel- and patch-level predictions are aggregated to a field label by majority vote (or, for probabilistic models, mean pooling). Computational cost across paradigms is reported in the Supplementary Material (Sec. S-C).

3.1. Classical Machine Learning

We automate feature extraction with the `xr.fresh` toolkit, computing a set of statistical and temporal descriptors (e.g., autocorrelation, linear trend, day-of-year of the seasonal maximum, quantiles, complexity) from each pixel’s 10-month profile, yielding a 174-dimensional per-pixel vector (Mann et al., 2026; Mann, 2025b); the full feature list is in the Supplementary Material (Sec. S-A, Table S1). Four baselines: logistic regression, RF (300 trees), LightGBM, and XGBoost (James et al., 2013; Breiman, 2001; Ke et al., 2017; Chen and Guestrin, 2016; Friedman, 2001) are trained on this representation, all of them tree- or margin-based with built-in feature selection or shrinkage; standard formulations are reproduced in the Supplementary Material (Sec. S-A). We also report soft-voting and stacked ensembles (RF/XGBoost/LightGBM, with logistic-regression meta-learner), an Optuna-tuned field-level XGBoost, and a SMOTETomek-resampled variant; the *Stacking* and *SMOTE Stacked* pair differs only in resampling, isolating the effect of class rebalancing. At the field level, pixel features are mean-pooled within each polygon before classification (Peña-Barragán et al., 2011).

3.2. Deep Learning Architectures

At the pixel level each observation is a 60-feature temporal sequence. Although the pixel models train on millions of pixels, pixels within a field are highly autocorrelated and share one label, so the number of statistically independent units is closer to the number of fields; field-wise splitting keeps evaluation honest at the field scale. We evaluate, as five-seed ensembles, a CNN-BiLSTM trained with weighted focal loss; TabNet (Arik and Pfister, 2021; Muruganantham et al., 2024), which treats the observations as tabular input and, at each decision step, uses an attentive transformer to produce a *sparse* feature mask via sparsemax, selecting a handful of features and ignoring the rest; the L-TAE (Garnot and Landrieu, 2020), which applies a single learned query over the temporal sequence; a standard multi-head Transformer encoder (Vaswani et al., 2017; Rußwurm and Körner, 2020); and TempCNN (Pelletier et al., 2019). Detailed equations for these standard models, and the field-level voting rule, are in the Supplementary Material (Sec. S-A). The Transformer is included as an informative dense-attention comparison: it increases attention capacity relative to L-TAE while retaining no explicit sparse feature-selection mechanism. Deep patch-based models (a multi-channel 2D CNN and a 3D CNN) tile each field into 100 m patches resampled to 128×128 and are likewise trained as five-seed ensembles; details in the Supplementary Material (Sec. S-A).

3.2.1. Sparse-Attention L-TAE (L-TAE-S)

To test whether a tree-like, sparse inductive bias can be grafted onto a temporal-attention encoder, we introduce L-TAE-S, which inserts a TabNet-inspired channel-selection gate before the attention heads (Fig. 2). For each of the $n_{\text{head}} = 16$ heads, a per-head linear projection of the embedded, positionally-encoded sequence ($d_{\text{model}} = 128$, key dimension $d_k = 8$) is passed through *sparsemax* (Martins and Astudillo, 2016) to produce a sparse, instance-dependent mask over embedding channels. Formally, for head h and time step $\mathbf{z}_t \in \mathbb{R}^E$,

$$\mathbf{m}_{h,t} = \text{sparsemax}(\mathbf{P}_{h,t} \odot \text{BN}(W_h \mathbf{z}_t)), \quad \tilde{\mathbf{z}}_{h,t} = \mathbf{m}_{h,t} \odot \mathbf{z}_t,$$

$$\mathbf{P}_{h+1,t} = \mathbf{P}_{h,t} \odot \max(\gamma - \mathbf{m}_{h,t}, 0), \quad \mathbf{P}_{1,t} = \mathbf{1};$$

head h then forms its keys and values from $\tilde{\mathbf{z}}_{h,t}$ before applying L-TAE attention. Because *sparsemax* projects onto the probability simplex, most channels of $\mathbf{m}_{h,t}$ are exactly zero, and the relaxation prior $\mathbf{P}_{h,t}$ (coefficient $\gamma = 1.5$) discourages successive heads from re-selecting the same channels. These two mechanisms are the sole source of sparsity. No explicit sparsity penalty is used, because an ℓ_1 penalty on a sparsemax mask is constant ($\|\mathbf{m}_{h,t}\|_1 = 1$) and has zero gradient. This single block, the only architectural difference from the L-TAE, grafts a tree-like sparse feature-selection bias onto the attention backbone while keeping its temporal modeling, mirroring TabNet’s sparsemax masks (Arik and Pfister, 2021). We evaluate L-TAE-S at both the pixel and field level as a five-seed ensemble under the same weighted-focal objective and field-aggregation protocol as the other temporal networks.

3.3. Performance Metrics

Because the classes are severely imbalanced, plain accuracy is misleading. We adopt macro-F1 ($F1_m$, the unweighted mean of per-class F1) as the primary metric, since it weights every crop equally and penalizes failure on the minority cereals where models differ most. We report three further metrics: Cohen’s Kappa; weighted F1 (wF1, the support-weighted mean of per-class F1); and cross-entropy (Xent), the field-level scoring metric of the AI4FoodSecurity challenge (AI4EO, 2021) from which the dataset is drawn (lower is better, computed on hard label predictions). Formal definitions are in the Supplementary Material (Sec. S-B).

3.4. Quantifying Uncertainty

We quantify two sources of variability. For sampling uncertainty in the finite holdout, we apply a nonparametric bootstrap over the 2,417 holdout fields: for each model we resample its per-field predictions with replacement $B = 10,000$ times and recompute macro-F1, reporting the 2.5th–97.5th percentile interval as a 95% confidence interval (CI). For between-model differences, we use a paired bootstrap: each replicate resamples the holdout fields once and scores both models on the same fields, giving 95% CIs for Δ macro-F1, $\Delta\kappa$, and Δ accuracy; an exact McNemar test on per-field correctness serves as a check. For

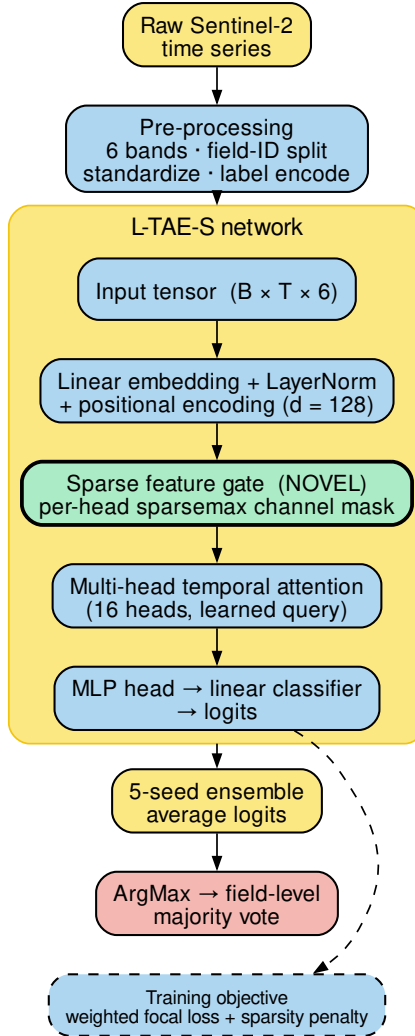


Figure 2: Architecture of the sparse-attention L-TAE-S. Before the multi-head temporal attention, a TabNet-style sparse feature gate (highlighted) applies a per-head **sparsemax** mask that selects a small subset of embedding channels, with a relaxation prior ($\gamma = 1.5$) discouraging heads from re-selecting the same channels. This block is the only architectural difference from the L-TAE; five seeds are ensemble and aggregated to field level.

238 optimization noise, the per-seed standard deviation of OOR macro-F1 across the
239 five ensemble members is 0.015–0.037 for the headline temporal models, which is

240 why we report five-seed ensembles throughout. Both sources are small relative
 241 to the ranking effects below: every field-bootstrap CI spans only $\approx \pm 0.02$ –0.03
 242 macro-F1, so single-model intervals separate only for differences of ≈ 0.05 or
 243 more. Specific comparisons are therefore tested with the paired bootstrap, which
 244 is more sensitive because field difficulty cancels.

245 4. Results

246 We evaluate every model under two protocols: 1) in-region field-wise val-
 247 idation and 2) out-of-region transfer (OOR) to the adjacent, non-overlapping
 248 holdout tile. All reported metrics are field-level. Pixel- and patch-level mod-
 249 els are aggregated to field labels, whereas field-level models are trained and
 250 evaluated on field-aggregated inputs

251 4.1. In-Region Validation Overstates Performance and Misranks Models

252 Table 1 reports both protocols for all models, ranked by OOR macro-F1,
 253 with the gap $\Delta = F1_m^{\text{OOR}} - F1_m^{\text{in-region}}$. Two facts stand out. First, *every*
 254 model loses accuracy under transfer, but unevenly: from -0.04 (field-level L-
 255 TAE-S) to -0.24 (L-TAE at the pixel level). Second, this unevenness reorders
 256 the models. In-region the dense temporal pixel nets cluster at the top (L-TAE
 257 0.78, Transformer 0.78, L-TAE-S 0.77, TempCNN 0.77, CNN-BiLSTM 0.72);
 258 a practitioner choosing by in-region F1 would pick from this cluster. On the
 259 holdout the cluster fractures: the Transformer drops to 0.58 and TempCNN to
 260 0.57 ($\Delta = -0.20$), CNN-BiLSTM to 0.51 ($\Delta = -0.21$), and the in-region leader
 261 L-TAE ($\Delta = -0.24$) to 0.54, below every pixel-level tree learner.

262 The winners under transfer are a different group. The best single model
 263 is the TabNet pixel ensemble ($F1_m = 0.60$, $\kappa = 0.54$), which also posts the
 264 lowest holdout cross-entropy (4.42) of any model, tied on macro-F1 by both
 265 sparse-attention L-TAE-S variants (0.60). The models with the smallest gaps
 266 are the sparse and linear learners: field-level L-TAE-S ($\Delta = -0.04$), logistic
 267 regression (-0.05), and the pixel-level tree learners. A small gap alone is not
 268 evidence of robustness, because it can also come from lower in-region accuracy:
 269 field-level L-TAE-S has the smallest gap largely because aggregation lowers its
 270 in-region macro-F1 from 0.77 to 0.64 while its holdout score stays at 0.60. We
 271 therefore read gaps together with holdout accuracy. On that joint view, L-
 272 TAE-S, a deep temporal attention network whose only change is a learned gate
 273 restricting each head to a small channel subset, ties TabNet for the best holdout
 274 accuracy at both the pixel and the field level, which supports our thesis. The
 275 paired bootstrap confirms which differences are real: the best transferers beat
 276 the dense pixel nets by wide, significant margins (TabNet over CNN-BiLSTM,
 277 $\Delta = 0.09$, 95% CI [0.07, 0.11]; over L-TAE, $\Delta = 0.06$, [0.04, 0.09]), so the
 278 inversion is not holdout noise; at the very top the leaders (TabNet and both
 279 L-TAE-S variants) are statistically indistinguishable (all 0.60; TabNet vs. L-
 280 TAE-S, $p = 0.73$). We therefore do not claim a single best model; we claim that
 281 a sparse-feature-selection group occupies the top of the OOR ranking while the
 282 dense temporal nets prized by in-region validation fall significantly below it.

Table 1: In-region versus out-of-region transfer performance of all twenty-three models.

Model	Level	Features	In-reg		Out-of-region (OOR)				
			$F1_m$	Δ	$F1_m$	95% CI	κ	wF1	Xent
TabNet (raw)	pixel	raw	0.73	-0.13	0.60	.58-.63	0.54	0.70	4.42
L-TAE-S	field	raw	0.64	- 0.04	0.60	.58-.62	0.52	0.69	4.77
L-TAE-S	pixel	raw	0.77	-0.17	0.60	.57-.62	0.53	0.70	4.63
L-TAE	field	raw	0.67	-0.07	0.59	.57-.62	0.47	0.67	5.51
Transformer	pixel	raw	0.78	-0.20	0.58	.55-.60	0.50	0.69	4.94
TempCNN	pixel	raw	0.77	-0.20	0.57	.55-.60	0.49	0.68	4.96
XGBoost	field	xr_fresh	0.70	-0.13	0.56	.54-.59	0.47	0.68	4.89
LightGBM	pixel	xr_fresh	0.62	-0.06	0.56	.54-.59	0.47	0.66	4.85
Logistic Regression	pixel	xr_fresh	0.61	-0.05	0.56	.54-.59	0.47	0.67	4.93
LightGBM	field	xr_fresh	0.68	-0.12	0.56	.53-.58	0.45	0.66	4.99
Voting	pixel	xr_fresh	0.69	-0.13	0.56	.53-.58	0.48	0.66	4.85
Random Forest	pixel	xr_fresh	0.65	-0.10	0.55	.52-.57	0.45	0.65	5.07
XGBoost	pixel	xr_fresh	0.64	-0.09	0.54	.52-.57	0.47	0.66	5.01
L-TAE	pixel	raw	0.78	-0.24	0.54	.51-.56	0.40	0.62	6.42
TempCNN	field	raw	0.67	-0.14	0.54	.51-.56	0.43	0.64	6.07
Stacking	field	xr_fresh	0.64	-0.11	0.53	.51-.56	0.48	0.67	5.32
CNN-BiLSTM	pixel	raw	0.72	-0.21	0.51	.49-.54	0.39	0.61	6.69
CNN-BiLSTM	field	raw	0.64	-0.14	0.51	.48-.53	0.33	0.54	7.86
3D CNN	patch	raw	0.70	-0.20	0.50	.48-.53	0.29	0.54	7.98
SMOTE Stacked	field	xr_fresh	0.69	-0.21	0.49	.46-.51	0.43	0.63	5.37
Multi-Channel 2D CNN	patch	raw	0.63	-0.18	0.45	.42-.47	0.30	0.54	7.62
TabNet (raw, field-avg)	field	raw	0.62	-0.19	0.43	.41-.46	0.31	0.56	7.35
TabNet (xr_fresh)	field	xr_fresh	0.51	-0.15	0.37	.34-.40	0.25	0.53	6.13

Ranked by OOR macro-F1. In-region uses field-wise validation; OOR uses the disjoint holdout tile (34S.20E.259N). Δ is the gap; the 95% CI is the percentile bootstrap on OOR macro-F1 from 10,000 resamples of the 2,417 holdout fields. Xent is the AI4FoodSecurity field-level scoring metric (lower is better). *Features* marks the automated **xr_fresh** statistics (174 features) versus raw reflectance. Bold marks the best value per column.

4.2. Pixel Versus Field Level: Aggregation as Variance Reduction

Several architectures were run at both the pixel and the field level (averaging per-pixel sequences within a field before classification). Table 2 shows that field-level aggregation narrows the gap of every *dense* temporal network (L-TAE $-0.24 \rightarrow -0.07$, TempCNN $-0.20 \rightarrow -0.14$, CNN-BiLSTM $-0.21 \rightarrow -0.14$), but it raises holdout accuracy only for L-TAE ($F1_m$ $0.54 \rightarrow 0.59$; paired $\Delta = 0.05$, 95% CI $[0.03, 0.08]$). For TempCNN ($0.57 \rightarrow 0.54$) and CNN-BiLSTM ($0.51 \rightarrow 0.51$) the gap closes mainly because in-region accuracy falls, so aggregation acts as a variance-reduction regularizer that trades in-region fit for stability rather than a reliable accuracy gain. The sparse-attention L-TAE-S follows the L-TAE pattern despite its channel gating: its gap shrinks from -0.17 to -0.04 while OOR $F1_m$ holds at 0.60, so the response to aggregation appears to depend on both the temporal backbone and the input granularity; the sparse gate does not eliminate the benefit of field-level aggregation. TabNet is the informative exception: aggregating its raw inputs to the field level *hurts* ($F1_m$ $0.60 \rightarrow 0.43$), because its robustness already derives from sparse feature selection and field-level training starves it of the per-pixel observation volume it exploits. The tree learners transfer well at either level, and their pixel $\rightarrow$ field

Table 2: Pixel- versus field-level out-of-region transfer.

Architecture	Pixel		Field		Field input
	$F1_m$	Δ	$F1_m$	Δ	
L-TAE-S	0.60	-0.17	0.60	-0.04	raw-avg
L-TAE	0.54	-0.24	0.59	-0.07	raw-avg
XGBoost	0.54	-0.09	0.56	-0.13	xr_fresh
LightGBM	0.56	-0.06	0.56	-0.12	xr_fresh
TempCNN	0.57	-0.20	0.54	-0.14	raw-avg
CNN-BiLSTM	0.51	-0.21	0.51	-0.14	raw-avg
TabNet	0.60	-0.13	0.43	-0.19	raw-avg

Architectures run at both granularities, ranked by field-level OOR $F1_m$. The *Field input* column gives the field-level input (**xr_fresh** statistics for the trees, field-averaged raw sequence for the deep models). Field aggregation narrows the gap for the dense temporal nets but raises their OOR $F1_m$ only for L-TAE, is roughly neutral for the tree learners, and degrades TabNet. Bold marks the best value per column.

change is essentially neutral under transfer (XGBoost $0.54 \rightarrow 0.56$). In short, aggregation stabilizes dense models but raises their transfer accuracy only for L-TAE; sparse, tree-like models do not need it.

4.3. Per-Class Performance and Persistent Confusions

Figure 3 breaks accuracy down crop by crop on the holdout for four representative models: the best transferer (TabNet), our sparse-attention L-TAE-S, its dense counterpart L-TAE, and CNN-BiLSTM (the weakest pixel-level temporal network on the holdout). The errors are highly uneven. TabNet and L-TAE-S classify the dominant lucerne/medics and the distinctive canola well ($F1$ 0.88 and 0.81; 0.87 and 0.76), L-TAE less so (0.77 and 0.67), but every model struggles with the cereals and small-grain grazing: even TabNet reaches only 0.55 on barley, 0.48 on wheat, and 0.30 on small-grain grazing. Adding the sparsity gate (L-TAE $\rightarrow$ L-TAE-S) on the same backbone improves every class, most on the easier ones (lucerne/medics $0.77 \rightarrow 0.87$, canola $0.67 \rightarrow 0.76$) and modestly on the cereals (barley $0.47 \rightarrow 0.50$, wheat $0.43 \rightarrow 0.47$, small-grain grazing $0.36 \rightarrow 0.40$). CNN-BiLSTM tracks L-TAE on most classes but is weakest on barley (0.33), so the dense networks lose accuracy on easy and hard classes alike, not only on minority classes. The wheat-barley confusion is consistent with their spectral overlap and appears across every model. It appears that errors are also spatially clustered and do not degrade with distance. Close to the training tiles, these four models and field-level XGBoost each classify about 70% of fields correctly; however, in the south-central zone (about 8 to 16 km away) L-TAE and CNN-BiLSTM fall to 36% and 30% while TabNet and L-TAE-S stay near 70% (Supplementary Material, Sec. S-E, Fig. S7).

4.4. Data Efficiency Under Out-of-Region Transfer

The most transferable models are also the most data-efficient. We retrain a representative set on random 75%, 50%, and 25% subsets of the training fields

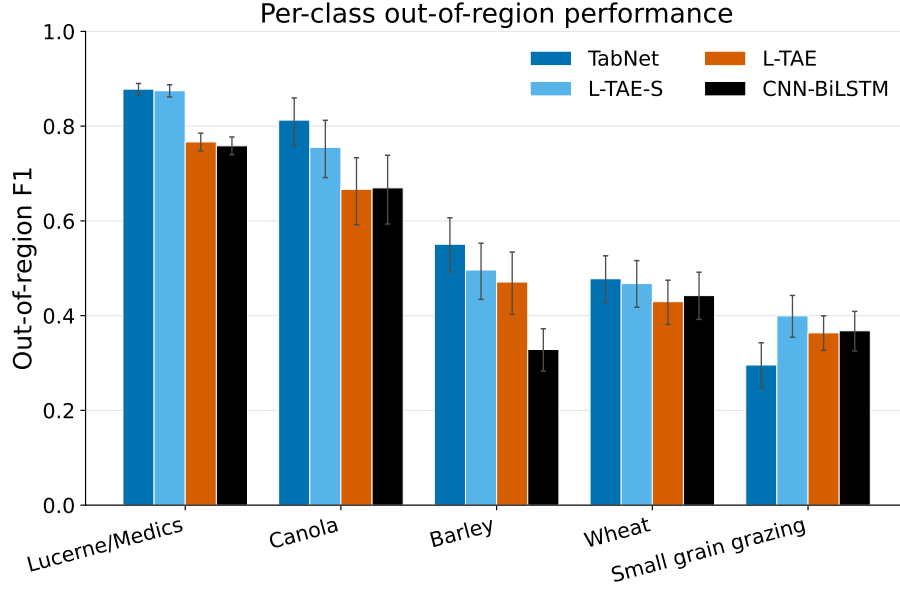


Figure 3: Per-class F1 on the holdout region for four representative models (TabNet, L-TAE-S, L-TAE, and CNN-BiLSTM). TabNet and L-TAE-S classify lucerne/medics and canola best; the cereals (barley, wheat) and small-grain grazing concentrate the errors. The dense L-TAE and CNN-BiLSTM also trail on the easier classes. Error bars are 95% bootstrap CIs over holdout fields.

and re-evaluate on the holdout; because a single subsample can be unrepresentative, we repeat each fraction over five independent draws and plot the mean (Fig. 4). Averaged this way, the deep temporal models are essentially flat from full data down to a quarter of the fields (L-TAE and L-TAE-S vary by under 0.02 macro-F1 across fractions), while the tree, linear, and TabNet baselines lose only ~ 0.03 – 0.05 , modest given a four-fold cut in training data. The per-fraction wiggles seen in any single split are within draw-to-draw sampling variance: L-TAE-S, whose sparse channel gate makes it the most sensitive to *which* fields are sampled, swings ± 0.06 macro-F1 at the 50% fraction (versus ≤ 0.02 for the robust learners), so the apparent 50% “dip” of any one draw is sampling noise rather than a data-efficiency effect. Holdout accuracy is therefore limited far more by out-of-region transfer than by training-set size, reinforcing that what governs holdout accuracy is the model’s inductive bias rather than the volume of training data. Because each subsample is drawn across the full training extent, this experiment varies the labeled training area (from ≈ 620 to ≈ 155 km² of fields) rather than the spatial extent of the training region.

4.5. Raw Reflectance versus `xr_fresh`

The best-transferring classical learners in Table 1 are exactly those fed `xr_fresh` statistics, so representation and model family are confounded there.

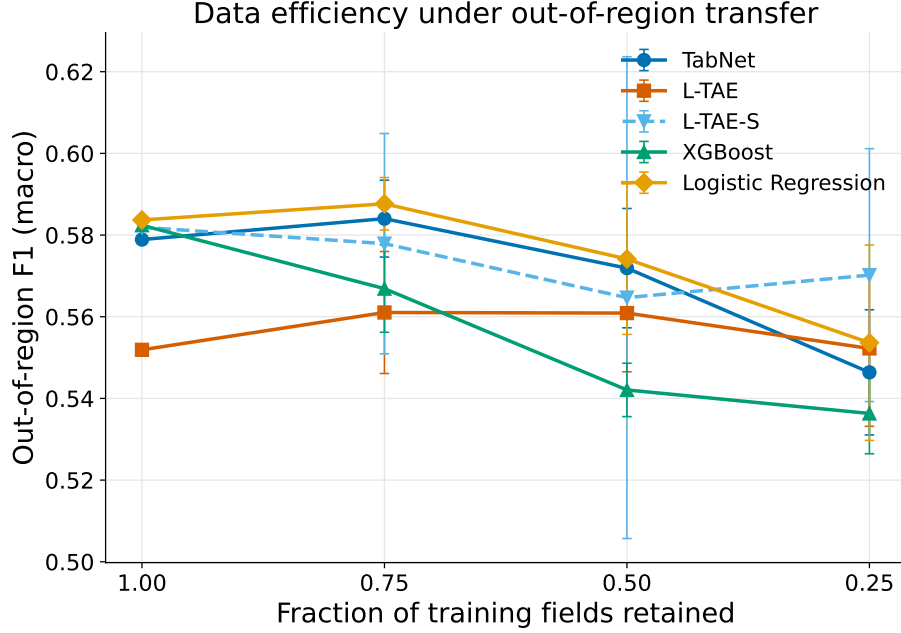


Figure 4: Data efficiency under out-of-region transfer: macro-F1 versus fraction of training fields retained, single training seed, each point the mean over five independent subsample draws (error bars ± 1 s.d.). The deep temporal models (L-TAE, L-TAE-S) are nearly flat to a quarter of the fields; the tree, linear, and TabNet baselines decline modestly. Per-fraction variation is within draw-to-draw sampling noise, largest for L-TAE-S (± 0.06 at 50%), so single-split “dips” are not real. Absolute values run ~ 0.02 – 0.05 below the five-seed levels of Table 1 because a single training seed is used.

352 To separate them, we trained a single XGBoost learner once on raw monthly
353 reflectance and once on the per-pixel `xr_fresh` statistics, holding the FID-wise
354 splits, preprocessing, and field-level mean pooling fixed (Table 3). The two rep-
355 resentations are essentially indistinguishable under transfer ($F1_m = 0.59$ raw
356 vs. 0.56 `xr_fresh`, a difference within the holdout bootstrap interval of Table 1;
357 equal weighted-F1 and κ), and raw reflectance is, if anything, marginally ahead
358 in-region as well. Field-level TabNet, run with the same configuration and
359 seeds on both inputs, points the same way: field-averaged raw sequences trans-
360 fer better than `xr_fresh` (OOR $F1_m = 0.43$ vs. 0.37 , non-overlapping bootstrap
361 intervals; Table 1). For these two learners, `xr_fresh` yields no transfer-accuracy
362 gain over raw reflectance. Its value is operational: a compact, fast-to-fit, agro-
363 nomically interpretable vector that trains in seconds (Supplementary Material,
364 Sec. S-C). The comparison does not extend to the deep temporal models, be-
365 cause `xr_fresh` reduces each time series to summary statistics with no time
366 axis for L-TAE, TempCNN, or CNN-BiLSTM to attend or convolve over. This
367 is, to our knowledge, the first controlled comparison of `xr_fresh` against raw
368 reflectance under out-of-region transfer.

Table 3: Raw reflectance versus **xr_fresh** for a single fixed XGBoost learner.

Representation	Split	$F1_m$	wF1	κ
Raw reflectance	In-region	0.73	0.80	0.71
xr_fresh	In-region	0.72	0.76	0.68
Raw reflectance	Out-of-region	0.59	0.69	0.50
xr_fresh	Out-of-region	0.56	0.69	0.50

Identical FID-wise splits, preprocessing, and field-level mean pooling; only the input representation differs. Metrics are field-level macro-F1 ($F1_m$), weighted F1 (wF1), and Cohen’s κ .

5. Discussion

The conclusion about which model is “best” depends almost entirely on whether evaluation is in-region or under out-of-region transfer, and a model’s transfer behavior is predictable from its inductive bias.

5.1. Why Tree-Like Inductive Bias Transfers

The clearest pattern is that transfer is strongly associated with inductive bias, and not simply with model family or attention capacity. Figure 5 groups every model by inductive bias and plots its OOR gap: linear models lose the least (mean $\Delta = -0.05$), tree-based models little (mean -0.11), the sparse-attention L-TAE-S sits essentially with the trees (mean -0.11 across its variants), while dense temporal/patch nets lose the most (mean -0.20) and SMOTE-augmented stacking is similarly fragile (-0.21). RF and gradient-boosted trees have been the remote-sensing workhorse for two decades not because they top in-region accuracy (dense nets often do) but because they survive when a model is applied to an unseen region (the operational case) (Belgiu and Drăguț, 2016; Maxwell et al., 2018). A tree decides one feature at a time, choosing the few features that matter for each split and ignoring the rest; this sparsity may reduce reliance on region-specific quirks of phenology, soil, and atmosphere that pull richer models off course. Independent corroboration spans settings: a 3D-CNN collapses under cross-sensor hyperspectral transfer while a sparse SVM holds (Pankajakshan et al., 2025); RF matches or exceeds a dense CNN-LSTM on label-scarce Ghana fields (Rustowicz et al., 2019); a maize model drops 16 points moving across East Africa (Jin et al., 2019); and bagged/boosted trees out-transfer an elaborate instance-reweighting scheme from the U.S. to northwest China (Mai et al., 2025). TabNet, the best transferer here, is a neural network built to behave like a tree: a small learned gate picks a handful of features at each step (Arik and Pfister, 2021), so it clusters with the trees and inherits their transferability. By contrast, CNN-BiLSTM, TempCNN, L-TAE, and the Transformer consume the whole temporal sequence at once with no feature selection. At the pixel level this leaves them fitting the training region tightly: they post the largest in-region-to-holdout gaps in the benchmark, with L-TAE dropping furthest ($\Delta = -0.24$). This is consistent with evidence that the dates such models attend to are themselves contingent on the training region (Obadić et al., 2022). Dense

attention is not intrinsically poor under out-of-region transfer: the pixel-level L-TAE loses 0.24 macro-F1, but the field-level L-TAE reduces this gap to 0.07 and raises holdout macro-F1 to 0.59 (Sec. 4.2), so it is the pixel-level dense models specifically that transfer worst.

The Transformer and L-TAE-S together separate attention capacity from feature selection on the same pixel-level task. The Transformer deploys far richer attention than L-TAE’s single query and does transfer better ($F1_m$ 0.58 vs. 0.54; paired $\Delta = 0.04$, 95% CI $[0.02, 0.06]$), so capacity is not irrelevant. Adding sparse feature selection to the smaller L-TAE helps more: L-TAE-S reaches 0.60 ($\Delta = 0.06$, $[0.04, 0.08]$; $\Delta\kappa = 0.13$) and matches or exceeds the Transformer (Δ macro-F1 0.02, $p = 0.06$; $\Delta\kappa$ and Δ accuracy 0.02, both significant). At the field level the gate’s macro-F1 gain over L-TAE is not significant (0.60 vs. 0.59), but κ and accuracy improve ($\Delta = 0.04$ and 0.05, both significant). L-TAE-S and TabNet, the two models that explicitly select features before computing with them, are the best transferers. Within the architectures tested, sparse feature selection is therefore associated with larger and more consistent transfer gains than added attention capacity; this is consistent with, but does not prove, feature selection as the property that matters most. The L-TAE-S experiment also shows that this is *not* a tree-versus-deep tradeoff: the same selection step used by tree splits can be grafted onto a dense architecture at the cost of a single block, and L-TAE-S ties TabNet for the best holdout accuracy. That parity is not accuracy alone: at the pixel level L-TAE-S reaches TabNet’s best-in-class holdout accuracy in roughly 40% of the per-seed training time (24 vs. 61 min), and its field-level variant is both faster and more accurate than any field-aggregated TabNet (Supplementary Material, Sec. S-C). The next generation of operational crop classifiers need not choose between the expressivity of deep learning and the transfer-robustness of trees.

5.2. Two Robustness Levers

Two caveats sharpen rather than weaken the story, and together they point to how we can choose the best models. First, it is *pixel-level* TabNet that wins; the field-aggregated TabNet variants are among the worst transferers (Table 1), because aggregation removes the per-pixel observation volume the sparse model exploits. Second, dense temporal attention transfers well *when* applied at the field level: L-TAE’s pixel-level gap shrinks from -0.24 to -0.07 under aggregation, and L-TAE-S ends with an even smaller gap (Table 2); for TempCNN and CNN-BiLSTM aggregation narrows the gap without raising holdout accuracy. These point to two distinct, partly substitutable directions. The first lever is built-in feature selection. The model itself picking a few features at a time (trees, TabNet, and now L-TAE-S); this is intrinsic robustness paid for at design time. The second is variance reduction imposed from outside via field-level aggregation, which averages away region-specific pixel noise before the classifier sees it. Which provides external robustness available even when the architecture is fixed. A model with the first lever gains nothing, and may lose observation volume, from the second; this is exactly why TabNet prefers pixels. These two properties can be combined. L-TAE-S adds a sparse gate to

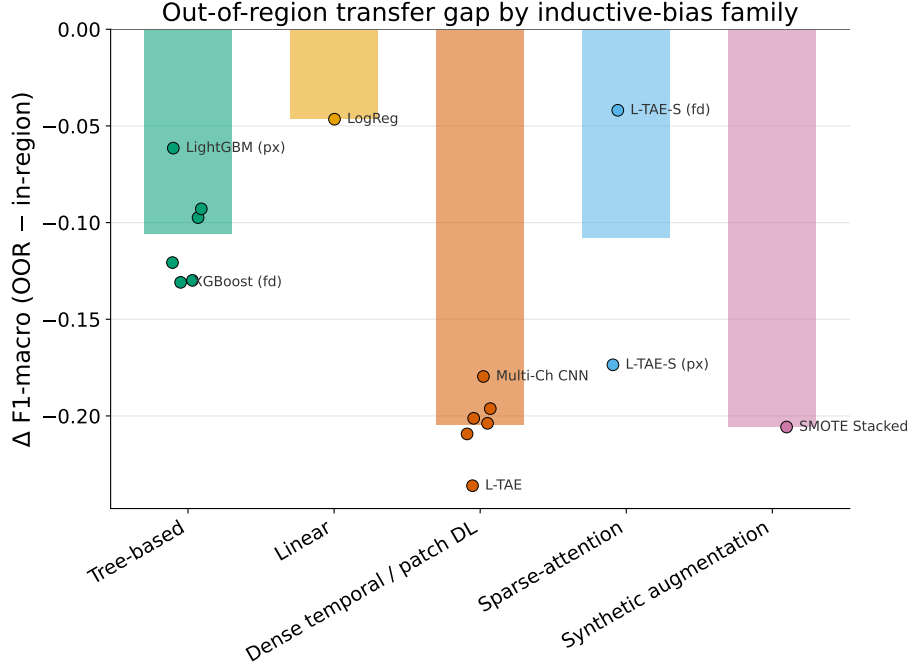


Figure 5: Out-of-region transfer gap (Δ macro-F1) grouped by inductive-bias family. Sparse/tree-like and linear models transfer (small $|\Delta|$); dense temporal/patch nets and synthetic augmentation overfit the training region. L-TAE-S, which adds sparse feature selection to a dense attention backbone, tracks the tree-based models. Bars are family means; points are individual models.

447 a temporal-attention backbone and then tolerates field aggregation: its holdout
 448 macro-F1 stays at 0.60 while per-seed training time falls from 24 min to 18 s.
 449 Practitioners can read this as a flowchart: if free to choose or modify the model,
 450 build feature selection in (lever 1); if constrained to an off-the-shelf dense ar-
 451 chitecture, aggregate to the field level (lever 2), which stabilizes transfer even
 452 when it does not raise accuracy. Two further controlled comparisons confirm
 453 the ranking is not an artifact of ensembling or imbalance handling; both shift
 454 absolute scores but never reorder the robust and fragile groups, with the full
 455 tables in the Supplementary Material (Sec. S-D). Seed-ensembling gives only
 456 a small, monotone lift that leaves the ordering intact. A single-seed L-TAE-S
 457 ($F1_m = 0.56$) already exceeds the five-seed CNN-BiLSTM and TabNet field
 458 ensembles (Table S4). And the imbalance-handling objective moves absolute
 459 scores but not the ranking: the robust L-TAE-S is almost invariant across five
 460 loss and sampler settings ($F1_m = 0.59\text{--}0.60$), whereas the fragile CNN-BiLSTM
 461 swings widely (0.41–0.56) and even at its best stays below L-TAE-S’s worst
 462 (Table S5).

5.3. Automated Features, Patch Models, and Persistent Difficulty

The classical pipelines paired with `xr_fresh` were not only robust but cheap, transferring as well as the best deep models (OOR $F1_m$ 0.55–0.56) while training in minutes rather than hours (Supplementary Material, Sec. S-C); the controlled comparisons for XGBoost and TabNet (Sec. 4.5) show that this value is operational rather than an accuracy edge over raw reflectance. Patch-based convolutional models did not exceed pixel-level models and transferred poorly (3D CNN $F1_m = 0.50$; multi-channel 2D CNN 0.45). We tested one configuration of each, so this should not be read as a property of patch CNNs in general. Several design factors may also contribute to it. At 10 m resolution a 100 m patch holds only $\sim 10 \times 10$ pixels, and upsampling to 128×128 adds interpolation rather than detail. After upsampling, the convolutional receptive fields span only one to two native pixels before global pooling or flattening, so little spatial structure is learned. The two models also fuse time differently: the 2D CNN stacks all band-month images as input channels, discarding temporal order, while the 3D CNN convolves across months. Lack of sparse feature selection may also play a role. Our experiments do not separate these factors; patch-size and architecture sweeps would be needed to do so. Finally, small-grain grazing, barley, and wheat are the hardest classes for every model (Fig. 3) and account for most accuracy lost under transfer; that even robust models struggle here, and that the dense networks trail on the easier classes as well, indicates the limitation is partly intrinsic spectral similarity and partly the transfer problem itself, not class imbalance alone. Much of this confusion is among the cereals themselves rather than with the other crops. Merging barley, wheat, and small-grain grazing into a single cereal class, without retraining, raises out-of-region accuracy from 0.73 to 0.84 for TabNet and from 0.71 to 0.83 for L-TAE-S, with the merged class reaching an F1 of about 0.77. Under the merged scheme the sparse models remain at the top of the ranking and the dense pixel networks at the bottom (Supplementary Material, Sec. S-F, Table S6).

5.4. Scope and Practical Relevance

The absolute accuracies reported here (out-of-region macro-F1 of at most 0.60) are too low for an operational crop map, and producing one is not our aim. The study is a controlled experiment on how model families behave under the low-information conditions typical of much of Africa: no labels in the target area, a modest labeled training set from a single season, and optical imagery only. Under these constraints the useful result is relative: which models lose the least when moved to unlabeled land, and how to choose among them when no target labels exist to check that choice. Three practical points follow. First, a model’s accuracy within the training region is a poor guide to how it will perform elsewhere, so when no labels exist in the target area, the safer choice is a model with built-in sparse feature selection, such as TabNet or L-TAE-S, or a dense model whose predictions are aggregated to the field level. Second, the crop-specific results are more useful than the overall score. Lucerne/medics, the dominant crop, and canola transfer well: TabNet and L-TAE-S reach out-of-region F1 of 0.87 to 0.88 for lucerne/medics and 0.76 to 0.81 for canola, so

these crops can be mapped with reasonable confidence in unlabeled neighboring areas. The cereals do not: barley (0.50 to 0.55), wheat (0.47 to 0.48), and small-grain grazing (0.30 to 0.40) remain confused with one another for every model, consistent with their overlapping spectral and phenological profiles. In this setting cereal maps are better reported as a combined cereal class, or improved with additional data such as SAR, than separated with optical time series alone. The sparse gate helps across all five crops (L-TAE to L-TAE-S), but it does not close the cereal gap. Third, the robust models are cheap to train: the classical pipelines need minutes and field-level L-TAE-S seconds per seed, which suits settings with limited computing resources. Adding SAR, multiple seasons, or a small number of target-region labels is the natural route from this benchmark toward operational accuracy.

5.5. Limitations

The study covers one growing season (2017) and one agroecological zone, with a holdout tile *adjacent* to the training tiles. Holdout fields lie 0 to 24 km from the nearest training field (median 11 km). The models are trained on ≈ 620 km² of labeled fields within a $\approx 25 \times 45$ km region and are used to predict in a $\approx 24 \times 24$ km tile. This is a relatively local shift, so the gaps we report most likely *understate* the degradation under cross-regional or cross-year transfer and are best read as a conservative lower bound. Cross-year and cross-ecozone transfer could not be tested, because the benchmark labels only these three tiles in a single season. Importantly, in South Africa, like much of Africa, training data is sparse if not non-existent. Multi-year, multi-region holdouts are needed to confirm that the inductive-bias ordering holds under larger shifts. Our conclusions therefore apply to out-of-region transfer between neighboring tiles at these distances and scales. Training-set size is the one scale we vary: the most transferable models keep near-full holdout accuracy on a quarter of the training fields (Sec. 4.4). The labels derive from a public dataset rather than our own campaign, and we use Sentinel-2 optical data only; a deliberate scoping choice that nonetheless limits the achievable ceiling for optically similar cereals, making SAR fusion a clear next step.

6. Conclusion

In-region validation, the protocol the crop-classification literature reports almost universally, misranks twenty-three Sentinel-2 classifiers when they are applied to an adjacent, non-overlapping holdout tile. Dense temporal networks that lead under conventional evaluation fall sharply under transfer, the in-region leader (L-TAE) dropping below every pixel-level tree learner, while parsimonious models such as tree ensembles, logistic regression, and a TabNet ensemble retain rank and accuracy. The common thread is the same property that made tree ensembles the remote-sensing default for two decades: a sparse, axis-aligned feature-selection bias that commits to a few robust signals and ignores the rest. TabNet inherits it through sparsemax masks. A Transformer with far greater

attention capacity transfers better than the lightweight L-TAE, but adding a sparse gate to L-TAE helps more, pointing to sparsity rather than capacity as the stronger lever. Grafting that lever onto an attention model (L-TAE-S) ties TabNet for the best transfer at a fraction of its training cost and, at the field level, keeps that accuracy while training in seconds per seed. Field-level aggregation is a second, partly substitutable lever: it narrows the transfer gap of dense temporal networks, raising holdout accuracy for L-TAE and L-TAE-S, but hurts TabNet. We therefore recommend the following: evaluate on an out-of-region holdout before declaring one architecture better than another; prefer models with a built-in feature-selection bias when mapping nearby unlabeled areas; and, if dense temporal models are required, aggregate their predictions to the field level. The conditions that make the Western Cape difficult are the conditions under which agricultural remote sensing must operate across much of the Global South, and evaluating models the way they are actually deployed, applied to regions where no labels were collected, is what will make remote sensing deliver where it matters most.

CRedit authorship contribution statement

Michael L. Mann: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Sairam Venkatachalam:** Software, Formal analysis, Writing – original draft, Investigation. **Disha Kacha:** Software, Formal analysis, Investigation, Writing – original draft. **Devarsh Sheth:** Software, Formal analysis, Investigation, Writing – original draft. **Ryan Engstrom:** Supervision, Methodology, Writing – review & editing. **Amir Jafari:** Methodology, Methodology, Supervision, Writing – review & editing.

Ethics statement

This article does not contain any studies with human participants or animals performed by any of the authors. The study is based solely on publicly available satellite imagery (Sentinel-2) and field-boundary crop labels, so no ethical approval was required.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The processed dataset supporting this study is openly available on IEEE DataPort at <https://dx.doi.org/10.21227/n69k-r458> (DOI: 10.21227/n69k-r458). The source code is available at https://github.com/mm1123/South_

588 Africa_Crop_Comp (Mann, 2025a). The study builds on the ground-reference
 589 crop labels of the Western Cape Department of Agriculture and Radiant Earth
 590 Foundation, “Crop Type Classification Dataset for Western Cape, South Africa”,
 591 v1.0, Radiant MLHub (DOI: 10.34911/rdnt.j0co8q), used under the CC-BY-4.0
 592 license; Sentinel-2 imagery was extracted via Google Earth Engine.

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